Comparison between the HyBIRT approach and EBRT-first sequence

HyBIRT (HDRIBT followed by IMRT)	EBRT followed by HDRIBT
Patients more receptive to the subsequent IMRT after HDRIBT induced tumor shrinkage.	Patients traumatized by EBRT toxicity may refuse further HDRIBT boost.
Easier to identify the GTV and tumor induration during the IBT applicator insertion.	Risk of "geographical miss" in HDRIBT.
IMRT isodose coverage can be "molded" to the earlier HDRIBT dose distribution.	Limited capability of HDRIBT to "mold" the dose distribution to conform the preceding EBRT.
Better sparring of parotid glands with the use of IMRT.	Limited capacity HDRIBT to compensate for IMRT dose non-uniformity within the target hence 3D-CRT technique is used.
Tumoricidal dose of radiation is delivered to the involved neck node/s and higk area with IMRT.	EBRT dose is limited to 40Gy – 54Gy in 20 – 27 to prevent overdose to subsequent HDRIBT overlap region.
May wait up to 6 months before subjecting the patient to salvage neck dissection.	Immediate salvage surgery for persistent neck node/s after sub-optimal dose of EBRT.
IMRT started immediately after the HDRIBT, reducing the risk of tumor repopulation.	Planned or salvage neck dissection after EBRT will prolong the OTT.

OC-0632 Local tumor control after brachytherapy impacts on survival in patients with uveal melanoma.

M. Gonzalez Cantero[1], M.M. Teja Ubach[1], I. Garrido Botella[1], M. Rodriguez Roldan[1], R. Rosel Aller[1], B. Deben Mendez[1], A. Ocanto Martinez[2], A. Escribano Uzcudun[1], E. Gonzalez del Portillo[1], I. Rodriguez Rodriguez[1], R. Morera Lopez[1]

[1]Hospital Universitario La Paz, Oncología Radioterápica, Madrid, Spain; [2]Hospital Universitario de Torrejón, Oncología Radioterápica, Madrid, Spain

Purpose or Objective

Uveal melanoma (UM) is the most common primary intraocular tumor in adults and treatment with episcleral-plaque brachytherapy (EPB) is the standard of care in medium-size tumors since the study COMS showed no difference in survival between EPB and enucleation.

The aim of this study is to assess the impact of local tumor control (LTC) of UM after EPB on survival.

Others objectives are to study disease-specific survival and overall survival at 5 and 10 years.

Materials and Methods

A retrospective observational study was conducted among 100 patients with UM treated with EPB with Iodine125 or Ruthenio106 from 2007 to 2013. Apex radiation dose was 85Gy with a dose rate of 50-125cGy/hour. LTC was evaluated periodically using ocular ultrasound, fundoscopy and retinography. Assessment of mortality and causes of death was carried out through the medical record of patients.

Results

LTC was achieved in 86 patients (86%). 5-year disease-specific survival was 94.9%. 5-year overall survival was 93%. 10-year disease-specific survival was 86%. 10-year overall survival was 80%. In the analysis of factors that might be involved in survival we obtained a significant result for LTC and disease-specific survival p: 0,018. There was no relationship between location (choroid, ciliary body, iris), gender, age, or eye-colour.